Bimodal Application Load: Iscandar vs Buddy

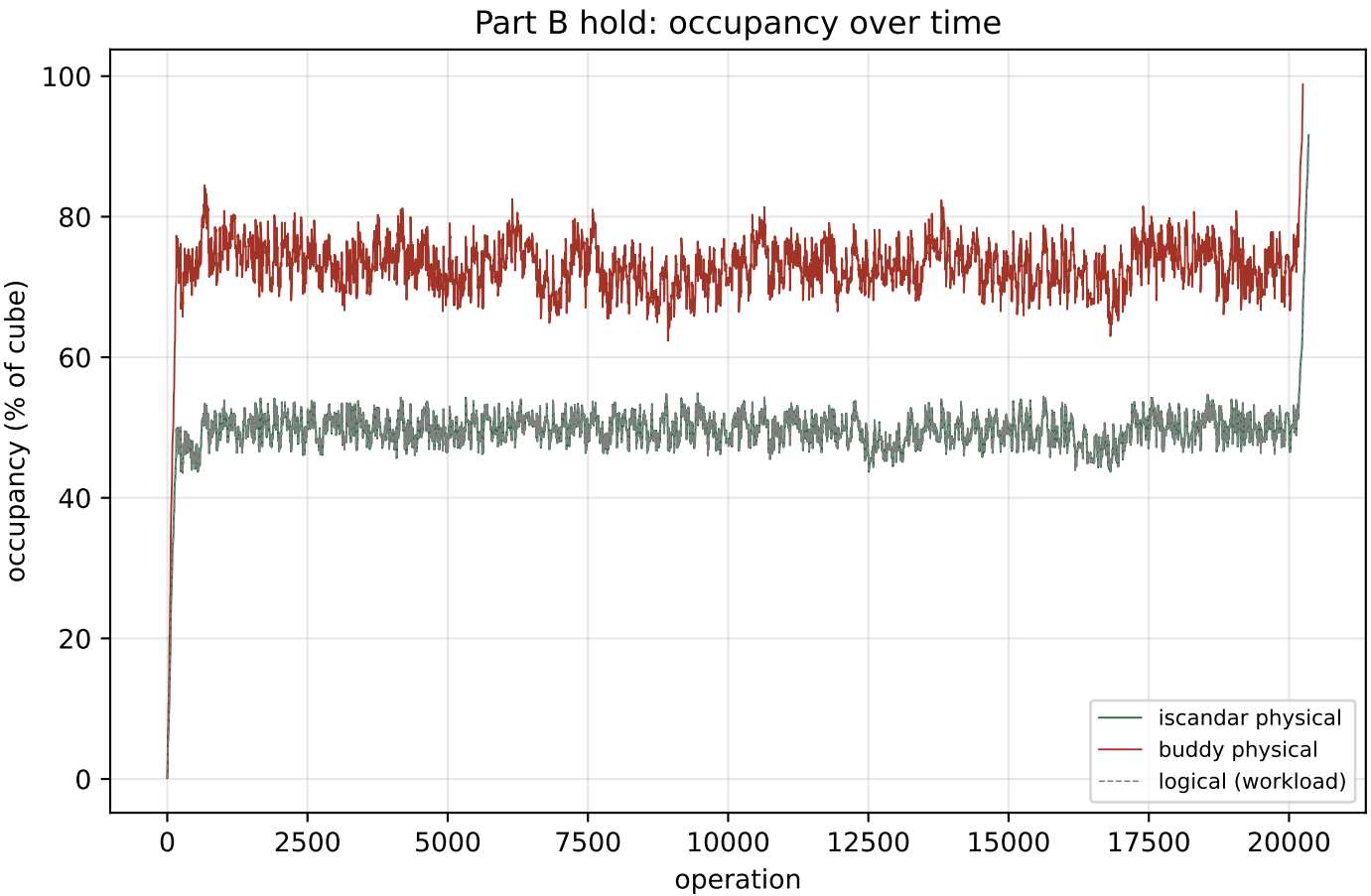
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Load: Bimodal Application Load (author's shelf,
  2026-10-26): a mixture - 90% of draws ~
  Normal(32, 10.5) clamped to 1..64, 10% ~
  Normal(512, 170.5) clamped to 65..1023. One
  application doing everything at once; the
  question is cross-mode interference.

Engines: the full Iscandar engine (all modes,
  exact fit) vs the classic binary buddy.
Arena: 32,768 sectors (512 KiB at 16 B/sector).
The same seeded stream drives both engines.

Part A - the sprint (fill to 85% logical, churn,
push; first failed allocation loses):
  buddy    stalled op 244 (ramp) at 99.96% physical
  iscandar stalled op 140 (ramp) at 42.59% physical
  margin 104 ops to buddy.

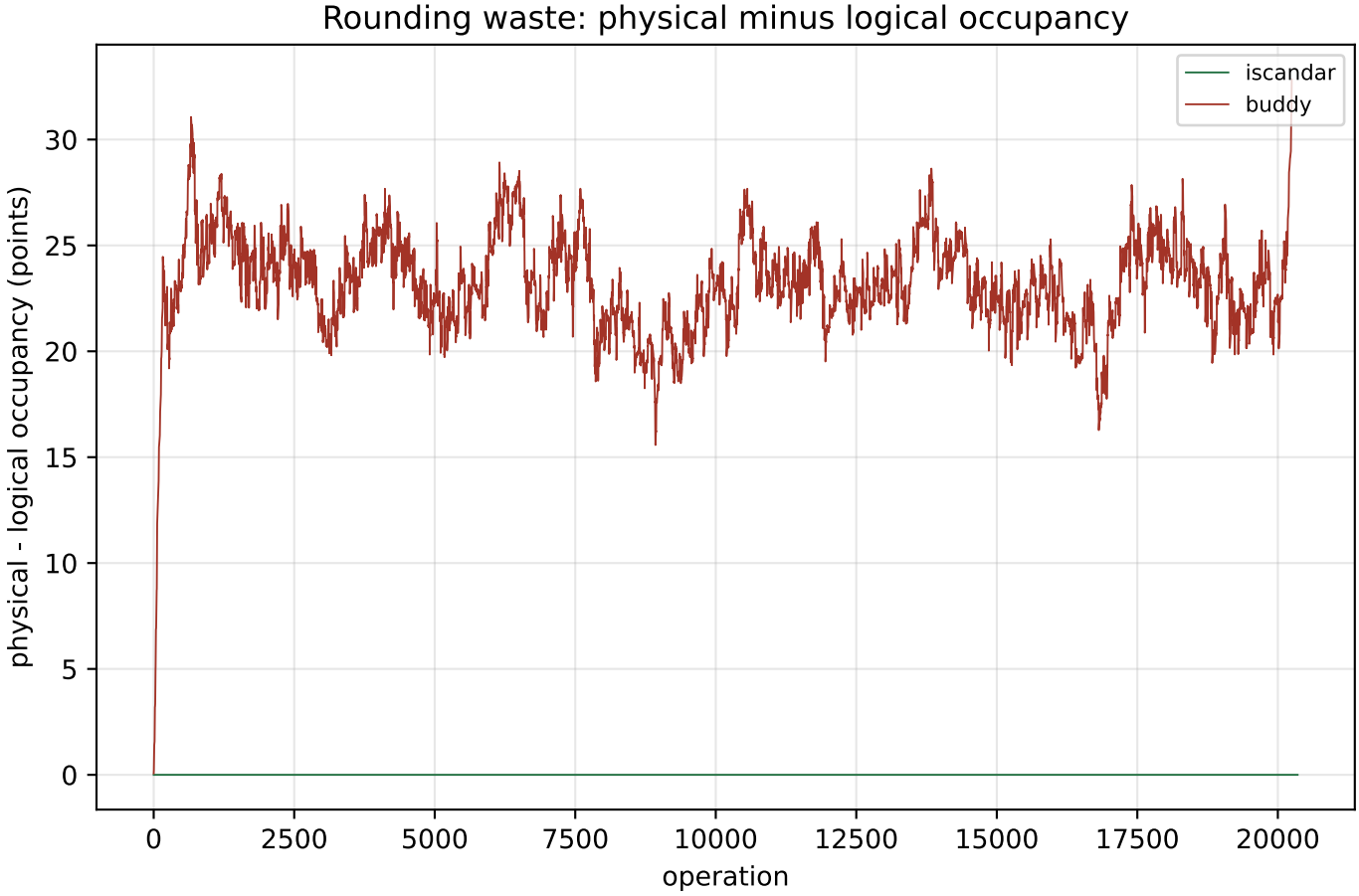
Part B - the steady-state hold (fill to 50%
logical, 20000 ops in a +/-2-point band, push):
  refusals during the hold: iscandar 2, buddy 2
  push stall: buddy at 98.86% physical, largest
            free block 256 sectors
  push stall: iscandar at 91.61% physical, largest
            free run 448 sectors

GammaCheck green in both parts (block checks
every op, full-ladder checks at boundaries).
```


Part B hold: occupancy over time
occupancy (% of cube)
operation
0
20
40
60
80
100
0
2500
5000
7500
10000
12500
15000
17500
20000
iscandar physical
buddy physical
logical (workload)

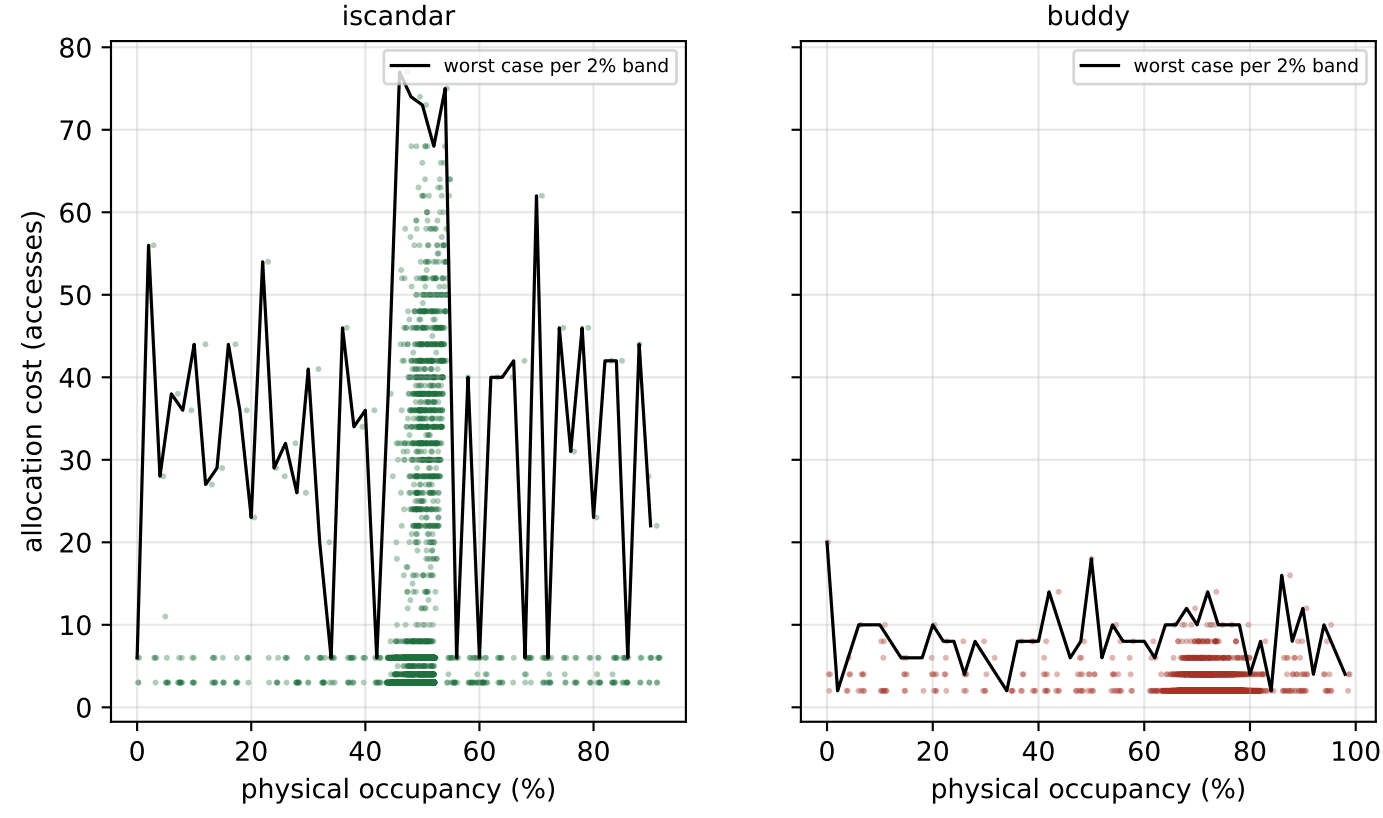
Commentary - occupancy over time

The gray dashed line is the workload itself, held at 50% logical. Iscandar's green line lies on top of it (mean 49.7% physical), because exact fit makes physical occupancy equal to logical occupancy. Buddy's red line rides above it: mean 73.1% physical, ranging 62.3 to 84.5%. The vertical distance between the red line and the workload is space Buddy has consumed without holding data.


Rounding waste: physical minus logical occupancy
iscandar
buddy
physical - logical occupancy (points)
0
5
10
15
20
25
30
0
2500
5000
7500
10000
12500
15000
17500
20000
operation

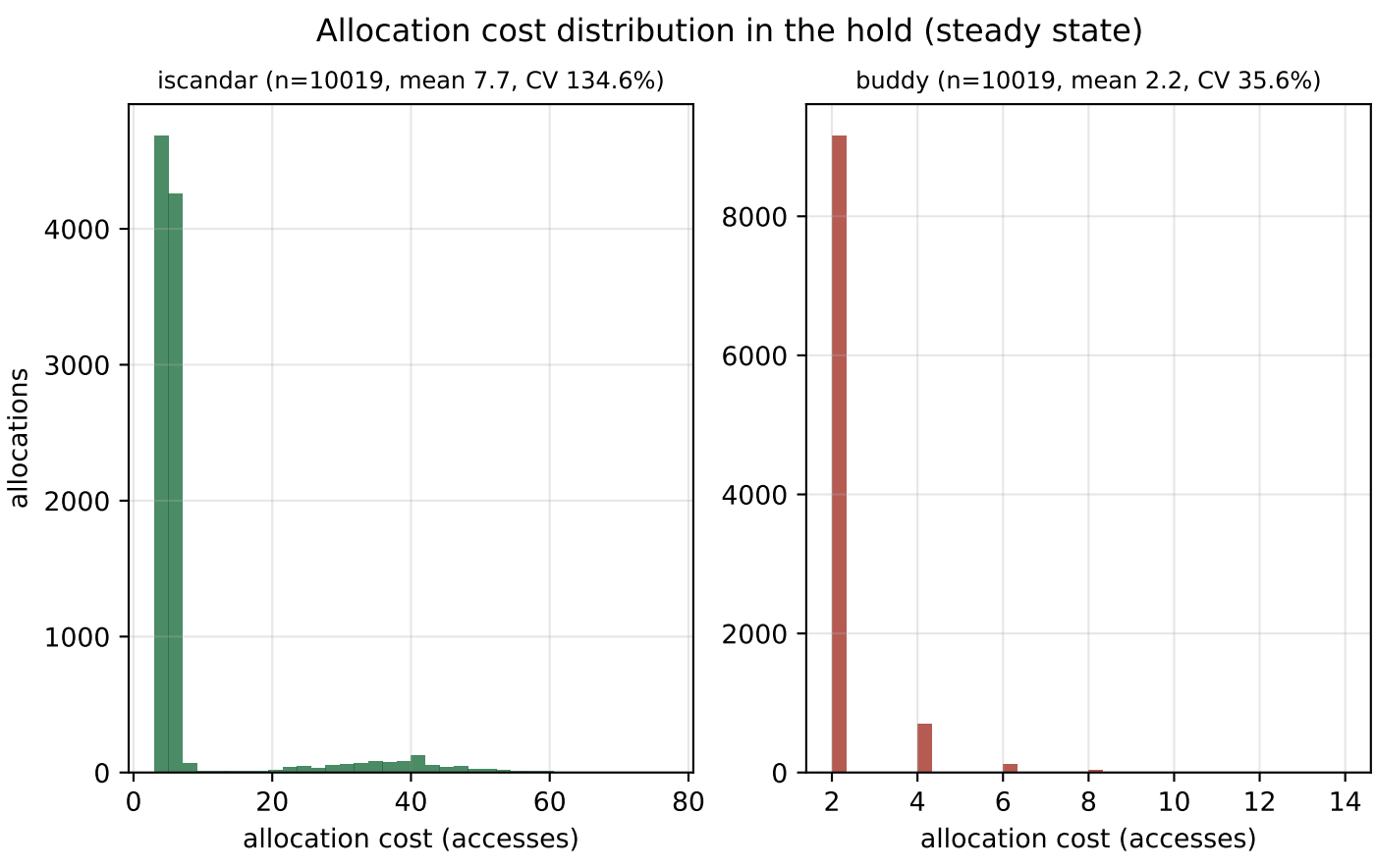
Commentary - rounding waste

Iscandar is a flat line on zero for the entire run. Buddy averages 23.3 points of pure waste and peaks at 31.1 - space that is neither data nor usable, only rounding.


Allocation cost against occupancy (hold + push; units differ per engine, see closing page)
iscandar
buddy
worst case per 2% band
worst case per 2% band
allocation cost (accesses)
physical occupancy (%)
physical occupancy (%)
0
10
20
30
40
50
60
70
80
0
20
40
60
80
0
20
40
60
80
100

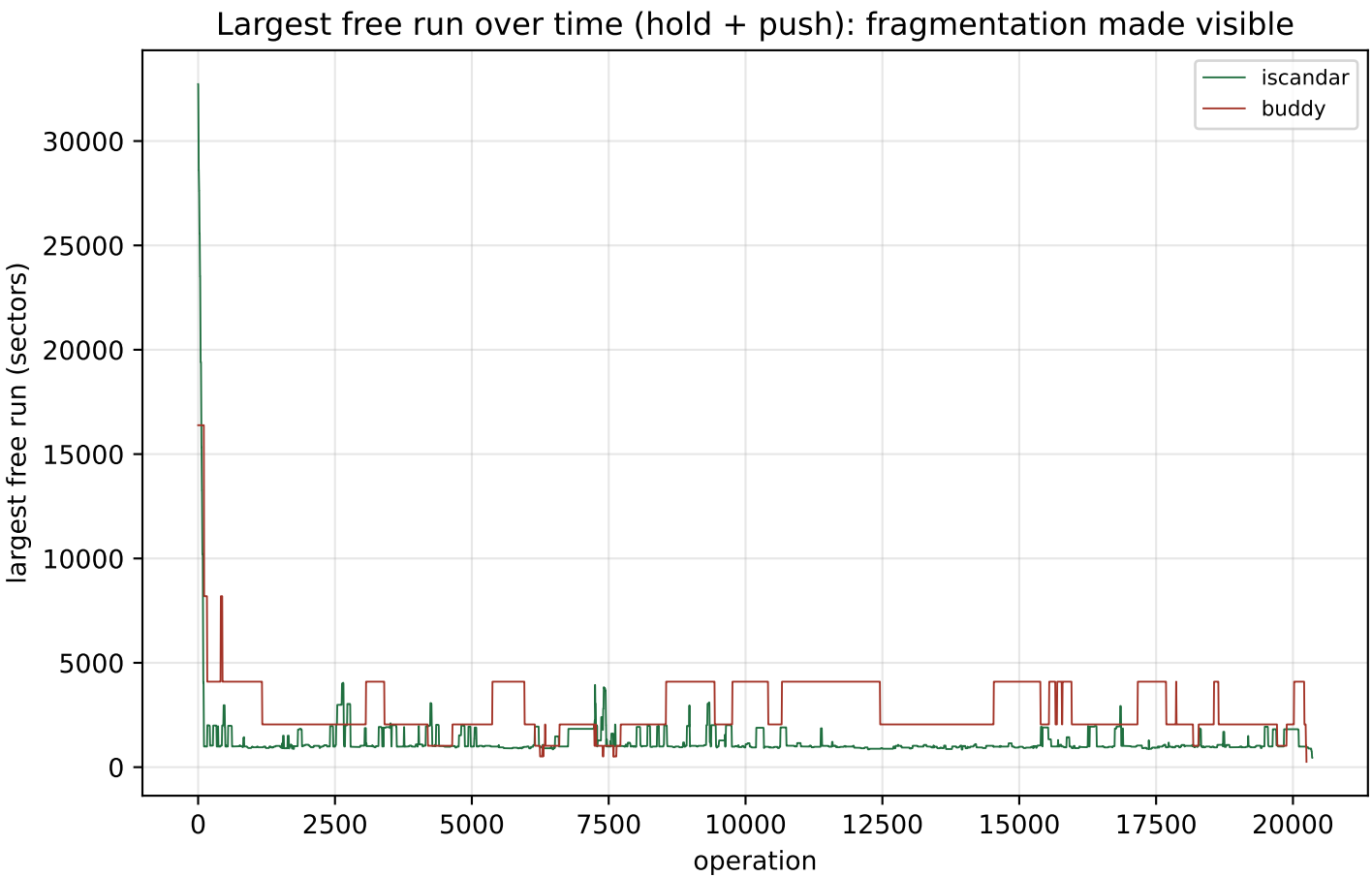
Commentary - cost against occupancy

Two panels with a black worst-case envelope. Iscandar's envelope stays between 6 and 77 touches across every occupancy band it visits; the price does not blow up as the cube fills. Buddy's envelope stays between 2 and 20 in its own tree-visit units (the levels are not comparable across panels; the shapes are). The readable point is the absence of a cliff: neither engine's worst case runs away with occupancy under this load.


Allocation cost distribution in the hold (steady state)
iscandar (n=10019, mean 7.7, CV 134.6%)
buddy (n=10019, mean 2.2, CV 35.6%)
allocations
0
1000
2000
3000
4000
6000
8000
0
20
40
60
80
2
4
6
8
10
12
14
allocation cost (accesses)
allocation cost (accesses)

Commentary - cost distributions

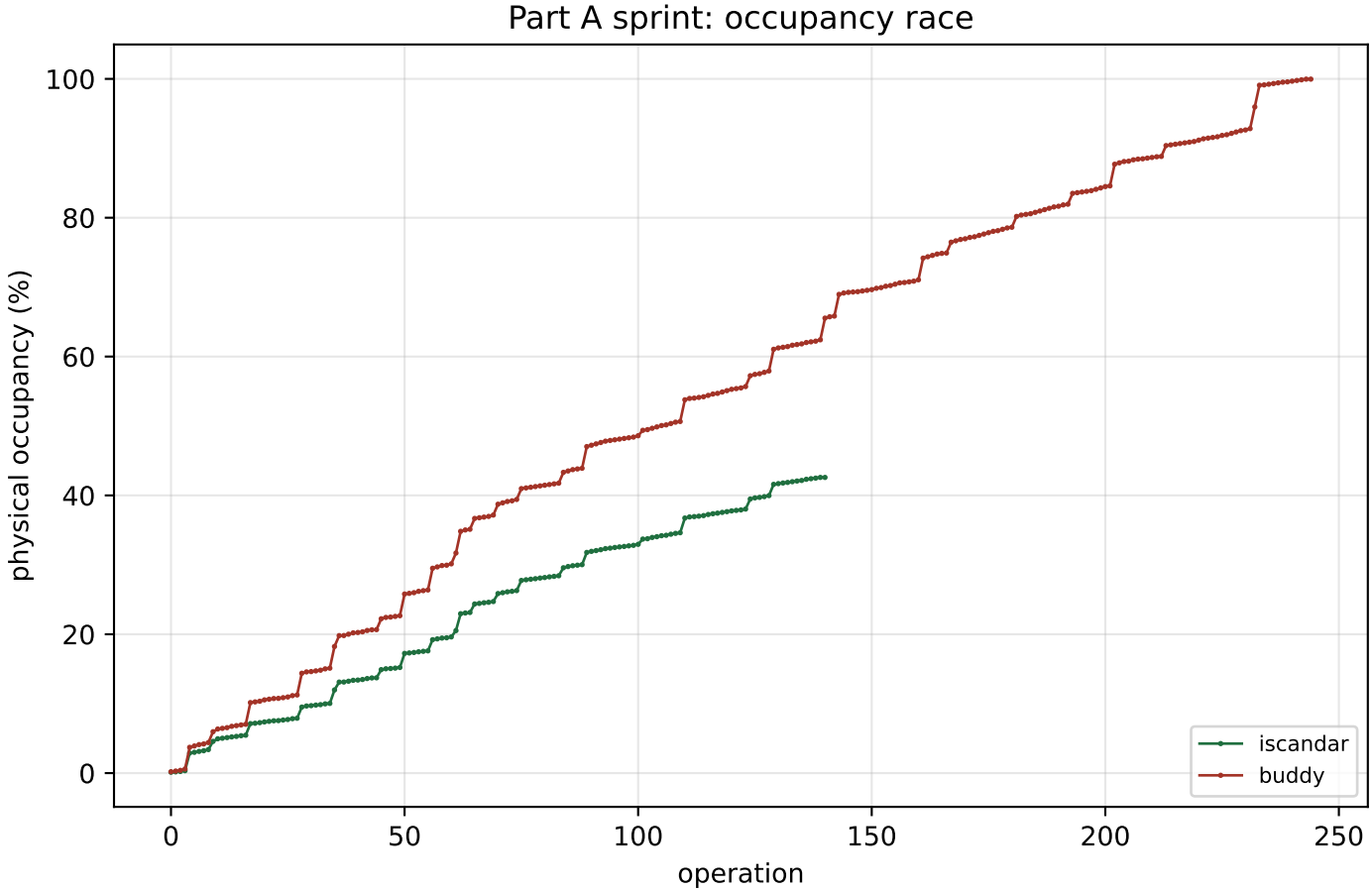
In the hold, Iscandar's allocation cost has median 6, with the middle half between 3 and 6 and the 90th percentile at 10. Buddy's has median 2, middle half 2 to 2, 90th percentile 2. The stability sigma on the closing page states the same shapes as a single number per series.


Largest free run over time (hold + push): fragmentation made visible
iscandar
buddy
largest free run (sectors)
30000
25000
20000
15000
10000
5000
0
0
2500
5000
7500
10000
12500
15000
17500
20000
operation

Commentary - largest free run

The fragmentation picture. Through the hold, Iscandar's largest contiguous free run has median 991 sectors; Buddy's largest free block has median 2048. At the push stalls, Buddy dies with a largest free block of 256 sectors at 98.86% physical, Iscandar with a largest free run of 448 at 91.61% physical.

Graph 5 is the one that makes the RTOS applications engineer stop and look. The occupancy graphs show what Buddy wastes; the largest-free-run graph shows what that waste costs - the size of the biggest promise each allocator can still keep.


Part A sprint: occupancy race
physical occupancy (%)
operation
0
20
40
60
80
100
0
50
100
150
200
250
iscandar
buddy

Commentary - the sprint

iscandar stalls first, at op 140 (42.59% physical); buddy continues alone and stalls at op 244 (99.96% physical). The margin is 104 ops to buddy.

The iscandar stall deserves a plain statement: it came on a request of 996 sectors with the cube only 42.59% full and a largest free run of 991 - 5 sectors short of the request. The space existed in total; it did not exist contiguously, because the other modes' traffic had salted every region of the cube. That is the cross-mode interference this profile exists to expose, and it cuts against Iscandar here: buddy wins this sprint.

Numbers and caveats

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Numbers (population statistics over successful ops)

sprint iscandar alloc: n=140 mean 8.73 var 133.33 CV 132.3% stability 0.8 sigma, range 3..56

sprint buddy alloc: n=244 mean 4.01 var 7.92 CV 70.2% stability 1.4 sigma, range 2..20

hold   iscandar alloc: n=10019 mean 7.74 var 108.40 CV 134.6% stability 0.7 sigma, range 3..77
hold   iscandar free:  n=9976 mean 4.55 var 26.11 CV 112.3% stability 0.9 sigma, range 2..34

hold   buddy alloc: n=10019 mean 2.22 var 0.62 CV 35.6% stability 2.8 sigma, range 2..14
hold   buddy free:  n=9977 mean 2.10 var 0.14 CV 17.7% stability 5.6 sigma, range 2..7

Counting and caveats

- Iscandar accesses are sector-word touches on
  the engine path: probe words in the search plus
  the commit's word touches, per the unified
  model in gaussian_race.py's header. Gate
  (counter) reads are a separate series and are
  not blended into accesses.
- Buddy accesses are its tree-work analog units
  (buddy.py), NOT the same physical units. Read
  across engines by shape, CV, and outcome;
  read levels and sigmas only within one engine
  against its own bound (the two-series
  discipline adopted 2026-10-26).
- Stability sigma is mean / SD (the reciprocal
  of CV): how many of its own standard
  deviations the mean cost sits above zero. It
  is a stability ratio, not a defect-rate
  sigma; a defect sigma needs a declared limit.
- Placement completeness (post-amendment): an
  Iscandar failure means no contiguous free run
  of the request existed, or the request
  exceeded all free space. It never fails facing
  an exact-fit hole.
- One seeded stream drove both engines in each
  part (seed 1). GammaCheck green throughout.
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